

WHITE PAPER

Soil Microbial Activity Monitor

A Low-Cost IoT-Based Device for Real-Time Soil Health Assessment

Using Raspberry Pi Pico W, Multi-Sensor Fusion, and Web-Based Analytics

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1.0.0

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Abstract

This white paper presents the **Soil Microbial Activity Monitor (SMAM)**, a low-cost, self-contained IoT device designed to provide farmers and agronomists with real-time insight into below-ground biological activity. Traditional soil testing requires laboratory analysis that is expensive, time-consuming, and provides only a point-in-time snapshot. The SMAM addresses this gap by continuously measuring five key soil and environmental parameters — **soil moisture, soil temperature, chamber humidity, soil pH, and CO₂ proxy gas concentration** — and fusing them into a simplified **Microbial Activity Index** classified as LOW, MEDIUM, or HIGH. The device is built around the **Raspberry Pi Pico W** microcontroller and communicates via WiFi to a Flask-based backend server, which hosts a farmer-friendly web dashboard accessible from any browser on the local network. This paper describes the system architecture, sensor selection rationale, signal processing methodology, activity classification algorithm, network implementation, and deployment guidelines. The SMAM is intended as an open, extensible platform for precision agriculture research and field deployment.

Key Metrics at a Glance	
Target Cost (BOM):	< USD 35 per unit
Measurement Cycle:	Every 12 seconds
ADC Averaging:	50-sample trimmed mean per reading
Sensors:	5 (MQ135, Soil Moisture, pH, DS18B20, DHT22)
Communication:	WiFi 802.11b/g/n (2.4 GHz) via Pico W
Dashboard Refresh:	Every 5 seconds (auto fetch)
Power Supply:	Li-Ion battery with USB charging
Operating Environment:	Outdoor agricultural field, IP-sealed top section

1. Introduction

1.1 Background and Motivation

Soil health is the foundation of sustainable agriculture. A single gram of healthy soil can harbour **one billion bacteria**, representing thousands of species collectively responsible for nutrient cycling, organic matter decomposition, nitrogen fixation, and disease suppression. The metabolic rate of this microbial community — often referred to as **microbial activity** — is the most sensitive and rapid indicator of soil biological health, responding to changes in moisture, temperature, pH, and organic carbon within hours to days.

Despite its significance, microbial activity is almost never monitored directly on farms. Conventional methods such as microbial biomass carbon analysis, enzyme assays, and CO₂ respiration tests require laboratory equipment, trained technicians, and turnaround times of days to weeks. This creates a critical observability gap: farmers apply fertilisers, irrigation, and tillage without feedback on how the soil ecosystem is responding.

The SMAM closes this gap with an affordable, field-deployable monitor that approximates microbial activity from readily measurable physical and chemical proxies, delivering real-time classification to any device with a web browser.

1.2 Problem Statement

The core challenge addressed by this innovation is:

Problem Statement
How can smallholder and commercial farmers obtain continuous, actionable information about
soil microbial health without access to laboratory infrastructure, using hardware costing
less than USD 35 and requiring no specialist technical knowledge to operate?

1.3 Innovation Scope

The SMAM innovation covers three integrated layers:

- **Hardware layer:** A cylindrical, three-section device with sensors distributed across soil, gas, and electronics zones
- **Firmware layer:** MicroPython firmware running on Pico W with sensor fusion, filtering, and WiFi telemetry
- **Software layer:** A Flask server and browser-based dashboard with farmer-friendly interpretation of raw sensor data

2. System Architecture

2.1 Physical Device Structure

The SMAM is housed in a cylindrical enclosure with three mechanically and functionally distinct sections. This vertical stratification mirrors the natural soil-air-electronics separation and enables appropriate environmental exposure for each sensor type.

Section	Environment	Contents / Function
Bottom Section	Open to soil	Soil moisture probe (capacitive analog), DS18B20 digital temperature sensor (OneWire), Al+Cu pH probe with LM358 op-amp buffer
Middle Section	Semi-closed gas chamber	MQ135 catalytic gas sensor (CO ₂ proxy), DHT22 digital temperature + humidity sensor; gas accumulates between measurements
Top Section	Fully sealed	Raspberry Pi Pico W, Li-Ion battery, TP4056 USB-C charging module, voltage regulator

2.2 Data Flow Architecture

The end-to-end data pipeline consists of four stages:

Stage	Description
1. Sensing	Five sensors output analog voltages or digital signals corresponding to physical/chemical parameters
2. Processing	Pico W firmware samples ADC channels (50 samples, trimmed mean), reads digital buses, computes derived metrics (gas rate, activity class)
3. Transmission	Pico W POSTs a JSON payload over WiFi to Flask server every 12 seconds
4. Presentation	Flask server stores latest payload in memory; dashboard HTML fetches /api every 5 seconds and renders values

2.3 Sensor Pin Assignment

GPIO / Bus	Sensor	Signal Type
GPIO26 (ADC0)	MQ135 gas sensor (via 1:1 voltage divider)	Analog — 0 to ~2.5V (divided from 5V AOUT)
GPIO27 (ADC1)	Soil moisture sensor	Analog — 0 to 3.3V (varies with soil water content)
GPIO28 (ADC2)	pH sensor via LM358 buffer	Analog — 0 to 3.3V (op-amp output)
GPIO2 (OneWire)	DS18B20 soil temperature	Digital — 1-Wire protocol, 4.7 k Ω pull-up to 3.3V
GPIO3	DHT22 chamber temperature + humidity	Digital — single-wire proprietary, 10 k Ω pull-up to 3.3V

3. Sensor Selection and Justification

3.1 Sensor Design Philosophy

Each sensor was selected to maximise the ratio of biological relevance to cost. Sensors were chosen to be individually available for under USD 5, use standard interfaces supported natively by MicroPython, and together provide a sufficient multivariate picture of conditions that determine microbial metabolic rate.

3.2 Sensor-by-Sensor Rationale

3.2.1 MQ135 — CO₂ Proxy (Microbial Respiration Indicator)

Biological significance: Soil microorganisms produce CO₂ as a byproduct of aerobic respiration during organic matter decomposition. The rate of CO₂ evolution from soil is the most widely used proxy for overall microbial metabolic activity in field ecology.

Sensor principle: The MQ135 is a tin dioxide (SnO₂) semiconductor sensor whose electrical resistance decreases upon exposure to reducing gases including CO₂, NH₃, and volatile organic compounds. While not selective for CO₂ alone, it provides a valid proxy in agricultural field conditions where the dominant gas source is microbial respiration.

Implementation note: The MQ135 operates on 5V and its AOUT can reach 5V, which exceeds the Pico W ADC maximum of 3.3V. A 1:1 resistor voltage divider (two 10 kΩ resistors) halves the output to a maximum of 2.5V, providing a safe margin. A 24-hour burn-in period is required before calibration.

3.2.2 Capacitive Soil Moisture Sensor

Biological significance: Soil water content directly controls microbial activity by governing solute transport, oxygen diffusion, and the water potential experienced by microbial cells. The Optimum moisture range for most soil bacteria is 40–70% of field capacity.

Sensor principle: Capacitive sensors measure the dielectric constant of the surrounding medium, which changes significantly with water content. Unlike resistive probes, they do not corrode and provide stable readings over years of field use.

3.2.3 Al+Cu pH Probe with LM358 Buffer

Biological significance: Soil pH determines the availability of nutrients, the speciation of toxic metals, and the composition of the microbial community. The optimal pH range for most soil bacteria is 6.0–7.5; outside this range, diversity and overall activity decline sharply.

Sensor principle: Aluminium and copper electrodes inserted into moist soil generate a millivolt-scale potential difference proportional to H⁺ ion activity. The LM358 operational amplifier in voltage-follower (unity-gain buffer) configuration prevents the high-impedance electrode from being loaded by the low-impedance ADC input, preserving signal integrity.

3.2.4 DS18B20 — Soil Temperature

Biological significance: Soil temperature is the primary determinant of the rate constant of all enzymatic reactions in microbial cells. Below 10°C most mesophilic bacteria enter dormancy; above 38°C thermal denaturation begins. The optimal range is 15–30°C.

Sensor principle: The DS18B20 is a digital OneWire temperature sensor with $\pm 0.5^\circ\text{C}$ accuracy over 0–70°C and 12-bit resolution. It is factory-calibrated and requires no additional circuitry beyond a 4.7 k Ω pull-up resistor.

3.2.5 DHT22 — Chamber Temperature and Humidity

Biological significance: The gas chamber humidity and temperature are needed to normalise the MQ135 reading (gas sensor resistance is temperature and humidity dependent) and to provide context for gas accumulation rate calculations. They also indicate the microclimate experienced by soil surface organisms.

Sensor principle: The DHT22 uses a capacitive humidity element and a thermistor, outputting calibrated values via a single-wire digital protocol with $\pm 2\%$ RH and $\pm 0.5^\circ\text{C}$ accuracy.

4. Signal Processing and Measurement Methodology

4.1 ADC Characteristics and Noise Sources

The Pico W incorporates a 12-bit successive-approximation ADC with an effective number of bits (ENOB) of approximately 8.7 bits due to internal reference noise and power supply coupling. In an agricultural field environment, additional noise sources include:

- 50/60 Hz mains interference from nearby electrical equipment
- Thermal noise in the LM358 op-amp and connecting cables
- MQ135 heater current transients (the MQ135 draws ~150 mA on the heater circuit)
- Soil ionic conduction creating variable electrode impedances

4.2 Trimmed-Mean Averaging Algorithm

To mitigate these noise sources, the firmware collects 50 ADC samples per reading with a 500 μ s inter-sample delay (total acquisition window: ~25 ms per sensor channel). The samples are sorted and the top and bottom 10% (5 samples each) are discarded before computing the arithmetic mean of the remaining 40 samples. This trimmed-mean approach provides:

- Spike rejection: transient electromagnetic interference affects only a few samples and is eliminated by the trim
- Stable baseline: the trimming ensures that the mean is not pulled by outliers from sensor warm-up transients
- Computation efficiency: sorting 50 integers on the Pico W takes < 1 ms, adding negligible overhead

4.3 Physical Parameter Conversion

4.3.1 Soil Moisture (%)

The raw ADC value (0–65535 for 16-bit representation) is linearly interpolated between two calibration points: the dry-air value (no moisture) and the fully-submerged value. The mapping formula is:

Moisture Conversion Formula
$\text{moisture_pct} = (\text{DRY_RAW} - \text{raw}) / (\text{DRY_RAW} - \text{WET_RAW}) \times 100$
Where: DRY_RAW = ADC value in air (~52000)
WET_RAW = ADC value submerged in water (~20000)
raw = current trimmed-mean ADC reading
Result is clamped to 0-100%

4.3.2 Soil pH

The LM358 buffer output is converted to a voltage ($V = \text{raw} / 65535 \times 3.3$), then to pH via a linear model centred on the neutral point:

pH Conversion Formula	
voltage	$= (\text{raw} / 65535) \times 3.3$
ph	$= \text{PH_OFFSET} - \text{PH_SCALE} \times (\text{voltage} - 1.65)$
Where: PH_OFFSET = 7.0 (pH at mid-voltage 1.65V)	
PH_SCALE = 3.5 (pH units per volt deviation)	
Result clamped to valid range 0.0-14.0	

4.3.3 CO₂ Proxy (ppm)

The MQ135 output (after voltage division) is linearly mapped from a calibrated raw ADC range to a ppm range:

Gas ppm Conversion Formula	
ppm	$= \text{PPM_MIN} + (\text{raw} - \text{RAW_MIN}) / (\text{RAW_MAX} - \text{RAW_MIN}) \times (\text{PPM_MAX} - \text{PPM_MIN})$
Where: RAW_MIN = 5000 (clean outdoor air baseline)	
RAW_MAX = 50000 (high-activity soil reading)	
PPM_MIN = 400 (atmospheric CO ₂ baseline)	
PPM_MAX = 5000 (elevated field reading)	

4.4 Gas Accumulation Rate

The gas accumulation rate is computed as the first-order finite difference of the CO₂ proxy ppm over the elapsed time since the previous measurement: **gas_rate = (current_ppm – previous_ppm) / elapsed_seconds**. This is stored in persistent firmware state variables and represents the rate of CO₂ evolution from the soil into the semi-closed gas chamber in units of ppm/second. A positive and rising gas_rate is the most direct proxy available for active microbial respiration.

5. Microbial Activity Classification Algorithm

5.1 Scientific Basis

The classification model is grounded in established soil ecology. The three parameters most strongly correlated with microbial metabolic rate, as reported in peer-reviewed soil science literature, are: (1) water-filled pore space (proxied by volumetric moisture content), (2) soil temperature (Arrhenius kinetics apply across the mesophilic range), and (3) carbon substrate availability (proxied by CO₂ evolution rate). The SMAM encodes these relationships as a simple additive scoring rule to produce an interpretable, low-latency classification.

5.2 Scoring Rules

Parameter	Score 0 (Poor)	Score 1 (Fair) / Score 2 (Optimal)
Gas Rate (ppm/s)	< 0.1 ppm/s	0.1 – 0.5 ppm/s (Score 1) > 0.5 ppm/s (Score 2)
Soil Moisture (%)	< 25% or > 85%	25–40% or 70–80% (Score 1) 40–70% (Score 2)
Soil Temperature (°C)	< 10°C or > 38°C	10–15°C or 30–38°C (Score 1) 15–30°C (Score 2)

5.3 Classification Thresholds

Total Score	Activity Level
5 or 6	HIGH — Conditions are optimal; microbial community is highly active
3 or 4	MEDIUM — Conditions are acceptable; suboptimal factor present
0, 1, or 2	LOW — One or more limiting factors present; activity suppressed

5.4 Limitations and Future Extensions

The current model treats all three parameters as equally weighted contributors and assumes a linear relationship between score and activity. This is an intentional simplification for a prototype device. Future versions of the algorithm should incorporate:

- Temperature correction of the MQ135 resistance using the DHT22 chamber temperature reading
- Exponential temperature response (Q10 model) rather than step-function scoring
- Substrate-adjusted thresholds based on soil organic carbon type (measured externally)
- Machine learning calibration against laboratory microbial biomass carbon measurements

6. Firmware Architecture (Raspberry Pi Pico W)

6.1 Language and Runtime

The firmware is written in MicroPython, which provides a Python 3 compatible environment with a small memory footprint (~256 KB flash), direct hardware register access, and a rich standard library including native WiFi, ADC, OneWire, and DHT drivers. This choice makes the code readable, modifiable by non-embedded engineers, and deployable without a C toolchain.

6.2 Module Structure

Module / Function	Responsibility
<code>read_adc_average()</code>	50-sample acquisition, sort, trim, mean for any ADC pin
<code>read_moisture_percent()</code>	ADC read + linear interpolation to moisture %
<code>read_ph()</code>	ADC read + voltage conversion + pH calculation
<code>read_mq135_ppm()</code>	ADC read + ppm linear interpolation
<code>read_soil_temperature()</code>	DS18B20 OneWire trigger + 750ms conversion wait + read
<code>read_dht22()</code>	DHT22 single-wire measurement with error handling
<code>compute_gas_rate()</code>	Stateful finite-difference rate calculation
<code>classify_activity()</code>	Additive scoring model returning LOW/MEDIUM/HIGH
<code>connect_wifi()</code>	STA mode WiFi connection with 20-attempt retry loop
<code>send_data()</code>	urequests HTTP POST with JSON payload and error handling
<code>main()</code>	Main loop: read → compute → classify → send → sleep

6.3 Fault Tolerance

The firmware implements several fault-tolerance mechanisms:

- DS18B20 and DHT22 read failures return None; None values are passed to the classifier which skips that parameter's score contribution
- WiFi disconnection is detected before each send cycle; automatic reconnection is attempted
- HTTP POST failures are caught and logged; the main loop continues to the next cycle
- ADC values are clamped to calibrated ranges before conversion to prevent out-of-range physical values

7. Network and Backend Implementation

7.1 WiFi Configuration

The Pico W connects as a WiFi station (STA mode) to a 2.4 GHz 802.11b/g/n access point using the onboard CYW43439 WiFi chip. SSID and password are stored as plain-text constants in the firmware. For production deployments, these should be stored in a separate configuration file or in flash memory outside the main firmware image.

IP Address Stability: The server IP is hardcoded in the Pico W firmware. To prevent connectivity failure after PC restarts, the server host should be assigned a static DHCP reservation in the router's admin interface (binding the host MAC address to a fixed IP). Alternatively, a static IP can be configured at the OS network interface level using NetworkManager (Linux) or Network Adapter Settings (Windows).

7.2 HTTP Communication Protocol

Data is transmitted via HTTP/1.1 POST to the /data endpoint every 12 seconds. The payload is UTF-8 encoded JSON with Content-Type: application/json. This choice of protocol was made for simplicity and broad compatibility; MQTT would be preferred in a production multi-device deployment for its lower overhead and QoS guarantees.

JSON Payload Schema	
{	
"moisture": 55.3,	// Soil moisture (0-100%)
"soil_temp": 22.1,	// DS18B20 reading (degrees C)
"humidity": 68.0,	// DHT22 chamber humidity (%)
"ph": 6.8,	// Computed pH (0.0-14.0)
"gas_ppm": 850.0,	// CO2 proxy concentration (ppm)
"gas_rate": 0.34,	// CO2 accumulation rate (ppm/s)
"activity": "HIGH"	// LOW / MEDIUM / HIGH
}	

7.3 Flask Backend Server

The backend is implemented as a single-file Python Flask application. It maintains the most recently received sensor payload in a Python dictionary in process memory. Three endpoints are exposed:

Endpoint	Method	Description
POST /data	POST	Validates and stores incoming JSON from Pico W; returns 200 OK or 400 with error message

GET /api	GET	Returns the latest stored sensor data as JSON; used by dashboard fetch()
GET /	GET	Serves the dashboard.html file with text/html content type

The server binds to 0.0.0.0:5000, making it accessible from all network interfaces, which is required for the Pico W on the same local network to reach it. In a production deployment, a reverse proxy (nginx) and process manager (gunicorn, systemd) should be added.

8. Web Dashboard

8.1 Design Principles

The dashboard is designed around a single overriding principle: a farmer with no technical background should be able to understand the soil status within three seconds of opening the page. This drives every design decision:

- The Microbial Activity level (LOW / MEDIUM / HIGH) is shown in the largest, most prominent element on the page, with colour-coded background (green / amber / red)
- Individual sensor values use large sans-serif numerals with their units clearly labelled
- Status badges below each value (e.g. "Optimal", "A bit dry", "Too cold") translate raw numbers into plain-language judgements
- A dynamic "Farmer Advice" section generates context-specific actionable text based on the current readings
- A progress bar at the top of the page provides a visual countdown to the next auto-refresh

8.2 Technology Stack

The dashboard is a single HTML file with embedded CSS and JavaScript. No build tools, frameworks, or external JavaScript libraries are required beyond two Google Fonts. This approach minimises dependencies, ensures the page loads instantly on slow local networks, and makes the code fully readable and modifiable by a developer.

Technology	Role
HTML5	Semantic page structure
CSS Custom Properties (variables)	Theming, consistent color palette
CSS Grid + Flexbox	Responsive sensor card layout
CSS Keyframe Animations	Refresh progress bar, activity banner flash, live indicator pulse
Vanilla JavaScript (ES2017)	fetch() API calls, DOM updates, badge logic, advice generation
Google Fonts (Syne + DM Sans)	Typography: display headings + body text

8.3 Auto-Refresh Implementation

JavaScript calls the /api endpoint every 5 seconds using the Fetch API. A cache-busting header (cache: "no-store") ensures the browser always retrieves fresh data from the server rather than a cached response. On each successful fetch, the DOM is updated in-place by writing to element textContent properties. A CSS flash animation is triggered on the activity banner when new data arrives to provide a visual confirmation of the update.

9. Hardware Implementation Guide

9.1 Bill of Materials

Component	Specification	Approx. Cost (USD)
Raspberry Pi Pico W	RP2040 + CYW43439 WiFi, 2MB flash	\$6.00
MQ135 Gas Sensor Module	5V, AOUT + DOUT, SnO ₂ element	\$2.50
Capacitive Soil Moisture Sensor	Analog output, 3.3–5V	\$2.00
DS18B20 Temperature Sensor	OneWire, TO-92 package, waterproof probe	\$2.00
DHT22 Humidity + Temp Sensor	±2% RH, ±0.5°C	\$3.50
LM358 Op-Amp (x1)	Dual op-amp, DIP-8, unity-gain buffer	\$0.30
Aluminum + Copper Electrodes	DIY from AL wire + Cu wire (5 cm each)	\$0.20
Resistors (4.7k, 10k ×2)	1/4W carbon film	\$0.10
18650 Li-Ion Cell	3.7V, 2000–3000 mAh	\$4.00
TP4056 Charging Module	USB-C, 1A, overcharge/discharge protection	\$1.00
MT3608 Boost Converter	3.7V → 5V for MQ135 heater	\$0.80
Cylindrical Enclosure + PCB	PVC pipe sections, proto-board	\$3.00
TOTAL		~\$25.40

9.2 Critical Wiring Notes

MQ135 Voltage Divider (MANDATORY): The MQ135 AOUT pin can reach 5V, which exceeds the Pico W ADC maximum of 3.3V and will permanently damage the ADC. Two 10 kΩ resistors in series from AOUT to GND, with the midpoint connected to GPIO26, create a 2:1 voltage divider (max 2.5V into ADC).

DS18B20 Pull-Up: The OneWire protocol requires a 4.7 kΩ pull-up resistor from the DATA line to 3.3V. Without this, the bus never reaches a logic-high level and all reads return None.

LM358 pH Buffer: Connect the LM358 in unity-gain configuration: non-inverting input (+) to the pH electrode output, inverting input (–) connected directly to the op-amp output. Power the LM358 from 3.3V (not 5V) to ensure the output swing stays within ADC range. The output feeds GPIO28.

Ground Plane: All sensors, the Pico W, and the LM358 must share a common ground. Floating grounds are the most common source of spurious ADC readings in field-deployed sensor systems.

10. Deployment and Field Operation

10.1 Installation Procedure

1. Flash MicroPython UF2 to the Pico W by holding BOOTSEL and connecting USB; drag UF2 to the mass-storage drive
2. Edit pico_firmware.py: set WIFI_SSID, WIFI_PASSWORD, and SERVER_IP to match your network
3. Copy pico_firmware.py to the Pico W as main.py using Thonny IDE (File → Save As → Raspberry Pi Pico)
4. On the server PC: install Flask (pip install flask), place server.py and dashboard.html in the same folder, run python server.py
5. Insert the SMAM device vertically into the soil so that the bottom section (sensor zone) is buried 10–15 cm deep
6. Connect power (USB-C to TP4056 module for charging, or fully charged 18650 for standalone)
7. Open a browser on any device on the same WiFi network and navigate to [http://\[SERVER_IP\]:5000](http://[SERVER_IP]:5000)

10.2 Calibration Schedule

Sensor	Calibration Procedure
Soil Moisture	Record ADC value in dry air and submerged in water; update MOISTURE_DRY_RAW and MOISTURE_WET_RAW
pH Probe	Immerse electrodes in pH 4.0 and pH 7.0 buffer solutions; adjust PH_OFFSET and PH_SCALE to match voltages
MQ135	Allow 24h burn-in; record clean-air baseline ADC value; update MQ135_MIN_RAW
DS18B20	Factory calibrated $\pm 0.5^{\circ}\text{C}$; no field calibration required
DHT22	Factory calibrated; no field calibration required

10.3 Battery Life Estimation

The dominant power consumer is the MQ135 heater circuit at approximately 150 mA on 5V (750 mW). The Pico W in active WiFi mode draws approximately 80 mA at 3.3V (264 mW). With a 3000 mAh 18650 cell (3.7V, 11.1 Wh), continuous operation endurance is approximately:

Battery Life Estimate	
Total power draw	$\approx 750 \text{ mW (MQ135)} + 264 \text{ mW (Pico W)} + 50 \text{ mW (other)} \approx 1064 \text{ mW}$
Energy capacity	$= 3000 \text{ mAh} \times 3.7\text{V} = 11,100 \text{ mWh}$

Estimated runtime = $11,100 / 1064 \approx 10.4$ hours continuous

For extended operation: put MQ135 on a MOSFET switch and power it only for 60s before each reading.

This duty-cycle approach can extend battery life to 3-5 days.

11. Validation and Testing

11.1 Unit Testing Without Hardware

The server and dashboard can be validated without the Pico W by sending simulated POST requests using curl (Linux/Mac) or Invoke-WebRequest (Windows PowerShell). Three representative test scenarios are recommended:

Scenario	Payload Values	Expected Activity
Optimal conditions	moisture:55, soil_temp:22, gas_rate:0.6, ph:6.8	HIGH
Dry and cool soil	moisture:20, soil_temp:8, gas_rate:0.05, ph:6.5	LOW
Moderate conditions	moisture:45, soil_temp:28, gas_rate:0.25, ph:7.1	MEDIUM

11.2 Known Limitations

- The Al+Cu pH probe has a limited lifespan (weeks to months) due to electrochemical corrosion; a commercial pH electrode is recommended for long-term deployments
- The MQ135 is cross-sensitive to ethanol, acetone, and other VOCs; in vineyards or near fermentation facilities, false-high readings are possible
- The linear ADC-to-pH conversion is an approximation; real electrode responses are logarithmic and require two-point calibration for accuracy better than ± 0.5 pH units
- In-memory data storage in the Flask server means all historical readings are lost on server restart; a SQLite database should be added for production use

12. Future Development Roadmap

Phase	Feature	Description
v1.1	Historical logging	SQLite database on server; time-series charts in dashboard
v1.2	MQTT telemetry	Replace HTTP POST with MQTT for lower overhead and multi-device support
v1.3	LoRa connectivity	Add SX1276 LoRa module for deployments beyond WiFi range (up to 5 km)
v2.0	Commercial pH electrode	Replace Al+Cu DIY probe with BNC-connector laboratory-grade electrode for ± 0.1 pH accuracy
v2.0	ML activity model	Train a regression model against laboratory microbial biomass carbon using field-collected data
v2.1	Solar charging	6V 1W solar panel with MPPT to enable indefinite field deployment
v3.0	Multi-node mesh	Distributed sensor nodes with central aggregation server and field-level mapping

13. Conclusion

The Soil Microbial Activity Monitor represents a significant step toward democratising soil biological monitoring. By combining five complementary sensors in a vertically stratified enclosure, applying robust signal processing techniques, and delivering results through a farmer-friendly interface, the SMAM makes real-time soil health intelligence accessible at a price point of under USD 35 per unit.

The multi-sensor fusion approach, while based on simplified linear models, provides a biologically grounded and empirically reasonable proxy for microbial metabolic activity. The architecture is deliberately open and extensible: every layer from firmware to dashboard can be modified independently, and the clear code structure enables rapid iteration and academic reproducibility.

The most significant innovation in this work is not any individual component, but the integration of sensor fusion, embedded signal processing, WiFi telemetry, and web-based presentation into a single cohesive system specifically designed for field deployment by non-technical users. This positions the SMAM as both a practical precision agriculture tool and a research platform for soil ecology instrumentation.

Summary of Innovations

1. Three-zone cylindrical sensor architecture separating soil, gas, and electronics environments
2. Al+Cu galvanic pH probe with LM358 buffer — a low-cost alternative to commercial electrodes
3. CO₂ proxy gas accumulation rate as a real-time microbial respiration indicator
4. 50-sample trimmed-mean ADC averaging optimised for agricultural EMI environments
5. Additive scoring microbial activity classifier grounded in soil ecology
6. Complete end-to-end open-source stack: MicroPython firmware + Flask server + HTML dashboard
7. Farmer-friendly dashboard with plain-language status badges and actionable advice

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Appendix A: Wiring Quick Reference

Pico W Pin	Sensor / Component	Notes
GPIO26 (ADC0)	MQ135 AOUT	Via 10k/10k voltage divider to GND
GPIO27 (ADC1)	Soil Moisture AOUT	Direct connection (sensor output $\leq$ 3.3V)
GPIO28 (ADC2)	LM358 output (pH buffer)	Op-amp powered from 3.3V rail
GPIO2	DS18B20 DATA	4.7 k Ω pull-up from DATA to 3.3V
GPIO3	DHT22 DATA	10 k Ω pull-up from DATA to 3.3V
3.3V (pin 36)	DS18B20 VCC, DHT22 VCC, Soil Moisture VCC, LM358 VCC	All low-power sensors
VBUS (pin 40)	MT3608 input (+5V out to MQ135 VCC)	Boosted from 3.7V Li-Ion via MT3608
GND (pin 38)	All sensor GNDs	Common ground — critical for ADC accuracy

Appendix B: Firmware Configuration Constants

Constant	Default Value / Description
WIFI_SSID	Your WiFi network name
WIFI_PASSWORD	Your WiFi password
SERVER_IP	IP address of PC running server.py (e.g. 192.168.1.100)
SERVER_PORT	5000 — Flask server port
SEND_INTERVAL_SEC	12 — seconds between data transmissions
ADC_SAMPLES	50 — samples per ADC reading (trimmed mean)
MOISTURE_DRY_RAW	52000 — ADC raw value in completely dry air (calibrate)
MOISTURE_WET_RAW	20000 — ADC raw value fully submerged (calibrate)
PH_OFFSET	7.0 — pH at mid-voltage (1.65 V)
PH_SCALE	3.5 — pH units per volt deviation from midpoint
MQ135_MIN_RAW	5000 — clean air baseline ADC value
MQ135_MAX_RAW	50000 — high-activity soil ADC value
MQ135_MIN_PPM	400 — CO2 ppm at min raw
MQ135_MAX_PPM	5000 — CO2 ppm at max raw

